RONALD KATENDE, PhD

Applied Mathematics | Mathematical AI | Scientific Computing | Mathematical Modelling

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Research profile

I study what information and computation are needed for reliable scientific inference. My work connects numerical analysis, mathematical AI, inverse problems and modelling through stability, approximation and error control. Applications include scientific learning, epidemics, climate resilience, economic change and conflict.

Research areas

- **Numerical analysis and scientific computing:** differential equations, numerical linear algebra, convergence, conditioning and error bounds.
- **Mathematical AI and learning theory:** neural approximation, physics-informed learning, information contraction and task-sufficient representations.
- **Inverse problems and scientific AI:** sparse sensing, identifiability, reduced-order models and uncertainty.
- **Mathematical modelling:** differential-equation, statistical and agent-based models in science and society.

Academic appointments

2026–present **Lecturer**, Department of Mathematics, Kabale University, Uganda.

2019–2026 **Assistant Lecturer**, Department of Mathematics, Kabale University, Uganda.

2016–2019 **Assistant Lecturer**, Uganda Christian University, Uganda.

Education

2025 **PhD in Mathematics**, Makerere University, Uganda. Scientific Computing and Machine Learning; thesis: *Numerical Analysis of Physics-Informed Neural Networks for Solving PDEs*.

2017 **MSc in Mathematics**, Pan African University / Jomo Kenyatta University of Agriculture and Technology, Kenya. Dissertation: *On the Location of a Free Boundary for American Options*.

2014 **BSc in Mathematics and Statistics**, Makerere University, Uganda. First Class Honours.

Peer-reviewed journal publications

Mathematical AI, approximation and scientific machine learning

1. **Katende, R.** (2026). Non-Asymptotic Stability and Consistency Guarantees for Physics-Informed Neural Networks via Coercive Operator Analysis. *Communications in Nonlinear Science and Numerical Simulation*, **159**, 109909. DOI.
2. **Katende, R.** (2026). Information contraction and stability under inverse-stable memory clocks in data-driven learning. *Research in Mathematics*, **13**(1), 2743449. DOI.

3. **Katende, R.** (2026). Rethinking Data-Efficient Artificial Intelligence for Low-Resource Settings. *Machine Learning with Applications*, **23**, 100796. DOI.
4. **Katende, R.**, Kasumba, H., Kakuba, G., & Mango, J. M. (2024). On the Error Bounds for ReLU Neural Networks. *IAENG International Journal of Applied Mathematics*, **54**(12), 2602–2611. Publisher PDF.
5. **Katende, R.**, Kasumba, H., Kakuba, G., & Mango, J. M. (2024). A Proof of Convergence and Equivalence for 1D Finite Element Methods and ReLU Neural Networks. *Annals of Mathematics and Computer Science*, **25**, 97–111. DOI.
6. Mango, J. M., **Katende, R.**, & Kasumba, H. (2025). Adaptive Error-Bounded Hierarchical Matrix Compression for Efficient and Accurate Physics-Informed Neural Networks. *Engineering Letters*, **33**(5), 1605–1622. Publisher PDF.

Scientific and societal modelling

7. Namugera, F., **Katende, R.**, Kurama, O., & Magero, J. M. (2025). Long-Range Epidemic Spreading with Aging in Structured Networks. *Physica A: Statistical Mechanics and its Applications*, **676**, 130841. DOI.
8. **Katende, R.** (2025). A Data-Efficient Framework for Analysing Structural Transformation in Low- and Middle-Income Economies. *Scientific Reports*, **15**, 34216. DOI.
9. **Katende, R.** (2026). Bridging the Macro–Micro Divide through a New Paradigm for Climate Resilience Assessment in Data-Scarce Regions. *PLOS Climate*, **5**(1), e0000662. DOI.
10. **Katende, R.** (2026). Modeling Resource Conflict and Inequality in Fragile Systems: A Spatial Agent-Based Framework with a Uganda Case Study. *PLOS ONE*, **21**(9), e0357283. DOI.
11. **Katende, R.** (2025). Cross-Country Comparative Analysis of Climate Resilience and Localized Mapping in Data-Sparse Regions. *Frontiers in Environmental Science*, **13**, 1495950. DOI.

Accepted conference work and presentations

- **Africa at NeurIPS 2026 (workshop):** *Compute What the Task Needs: Operational Decision Frontiers Beyond Raw Scale.* Accepted.
- **CITI 2027:** *No-Free-Necessity: Identifiability, Intervention Order, and Stable Counterfactual Certification in Learned Systems.* Accepted.
- **DeepMath 2026:** *When Can Depth Replace Width? A Sharp ReLU Depth–Width Frontier with Persistent State.* Poster accepted.
- **FLICS 2026, Valencia:** Two papers accepted for presentation.

Teaching and supervision

Numerical mathematics	Numerical Analysis; Advanced Numerical Methods (postgraduate); Numerical Linear Algebra; Linear Algebra; Differential Equations; PDEs.
Analysis and optimisation	Real and Functional Analysis; Linear Programming; Mathematical Modelling.
Statistics and programming	Probability and Statistics; Statistical Modelling; Data Analytics; Python, R and scientific-computing laboratories.

- Undergraduate and postgraduate teaching since 2016; typically at least 11 credit units per semester. Intensive Python training for MSc Statistics and month-long Python instruction for MSc Computational Mathematics at Makerere University.
- Two completed MSc supervisions, three additional MSc research projects and over 50 undergraduate research projects supervised or guided.

Current investigations and research prototypes

Ongoing research and prototypes; not reported as established results or deployed services.

- **Task-sufficient learning:** information limits, task-relative conditioning, computation and counterfactual certification.
- **Sparse-sensor inverse recovery:** sensor selection, full-field reconstruction and regime-dependent validation.
- **GridTwin Sentinel (prototype):** physical consistency, uncertainty and low-compute electricity forecasting.
- **AMMED (prototype):** multi-timescale memory for clinical early warning; transfer and false-alarm tests planned.

Research funding and academic visits

Research support: Kabale University Research and Innovation Fund, awardee (2025–2026); Government of Uganda research and innovation projects, co-investigator (2022–2024 and 2023–2025).

International academic experience

- University of Bergen, Norway: scientific machine learning and numerical analysis collaboration during doctoral studies.
- Area Science Park, Trieste, Italy: three-month research visit in sparse optimisation and data-limited computation (2025).
- Justus Liebig University Giessen, Germany: research visit and seminar (2025); University of Havana, Cuba: academic visit and seminar (2023).
- Other engagements: University of Milan, Italy; University of Rwanda.

Awards and professional service

Awards and fellowships: TWAS Research Award (2025); Computational Geometry Fellowship, University of Agder (2024); Mathematics for Africa Research Prize, European Mathematical Society (2023); CIMPA Fellowship (2023).

Academic service

- Mathematics representative, Faculty of Science Research Committee; Coordinator of Innovations and Exhibitions, Department of Mathematics, Kabale University.
- Curriculum development, including a proposed Computational Science and Industrial Analytics programme; National Council for Higher Education Innovation Exhibition.
- Journal reviewer: *Scientific Data*, *Neurocomputing*, *Machine Learning: Science and Technology*, *International Journal of Machine Learning and Cybernetics*, *International Journal of Data Science and Analytics*, *Discover Artificial Intelligence*, *Discover Sustainability*, among others.
- Conference reviewer: ITU Kaleidoscope; Africa at NeurIPS 2026.

Technical practice

Methods: Numerical and functional analysis, PDEs, numerical linear algebra, approximation, stability, conditioning, optimisation, inverse problems and scientific machine learning.

Software: Python, NumPy, SciPy, scikit-learn, PyTorch, MATLAB, R, Git, Linux and LaTeX; reproducible experiments and version-controlled workflows.